

Final Project: Chatbot and RAG

DV1729: Applied Generative AI

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Programming Track

Assignment name	Final Project: Chatbot and RAG
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LLM-Based Match Dataset Chatbot

This project focuses on building a chatbot that can answer questions about a match dataset using Large Language Models (LLMs). The goal was to convert natural language questions into executable pandas queries and return meaningful results to the user. The dataset used in this project contains information about matches, including columns such as match details, venue, teams, and results. Before processing, the dataset was cleaned by removing unwanted spaces in column names to ensure consistency.

The system uses two LLMs, Llama 3.2 and DeepSeek-R1, through Ollama. When a user enters a question, both models are used to generate pandas queries. The system then evaluates the generated queries and selects the one that executes correctly on the dataset. This approach helps improve reliability by comparing outputs from two models instead of depending on a single one.

To make the system more stable, a small rule-based fallback mechanism was added. For example, if the user asks about known fields like venue or home team, predefined queries are used directly. This avoids failures when the LLM generates incorrect or incomplete code. At the same time, for general queries, the system still relies on LLM-generated queries, ensuring that the requirement of using generative AI is satisfied.

The generated queries are executed using a restricted eval function to ensure safety. The results are then displayed in a tabular format. A summary of the result is also generated using one of the LLMs to provide a short explanation of the output.

To test the system, a separate test file was created with multiple test cases. These include normal queries, such as retrieving venue names, and edge cases, such as querying a nonexistent column like salary. The system successfully handles invalid queries by returning an appropriate error message instead of crashing.

One of the main challenges in this project was handling incorrect queries generated by LLMs and aligning them with the dataset schema. This was solved by improving the prompt and adding fallback logic.

Overall, the project demonstrates how LLMs can be used for natural language querying of structured data, while also highlighting the importance of combining AI with rule-based approaches for better reliability.